



Database Design Using the E-R Model

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Outline

- Extended E-R Features
- Entity-Relationship Design Issues
- Alternative Notations for Modeling Data
- Other Aspects of Database Design

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Design Phases

- Initial phase -- characterize fully the data needs of the prospective database users.
- Second phase -- choosing a data model
 - Applying the concepts of the chosen data model
 - Translating these requirements into a conceptual schema of the database.
 - A fully developed conceptual schema indicates the functional requirements of the enterprise.
 - Describe the kinds of operations (or transactions) that will be performed on the data.

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Design Phases (Cont.)

- Final Phase -- Moving from an abstract data model to the implementation of the database
 - Logical Design -- Deciding on the database schema.
 - Database design requires that we find a "good" collection of relation schemas.
 - Business decision -- What attributes should we record in the database?
 - Computer Science decision -- What relation schemas should we have and how should the attributes be distributed among the various relation schemas?
 - Physical Design -- Deciding on the physical layout of the database

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Design Alternatives

- In designing a database schema, we must ensure that we avoid two major pitfalls:
 - Redundancy: a bad design may result in repeat information.
 - Redundant representation of information may lead to data inconsistency among the various copies of information
 - Incompleteness: a bad design may make certain aspects of the enterprise difficult or impossible to model.
- Avoiding bad designs is not enough. There may be a large number of good designs from which we must choose.

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Design Approaches

- Entity Relationship Model
- Models an enterprise as a collection of *entities* and *relationships*
 - Entity: a "thing" or "object" in the enterprise that is distinguishable from other objects
 - Described by a set of *attributes*
 - Relationship: an association among several entities
 - Represented diagrammatically by an *entity-relationship diagram*:
- Normalization Theory
- Formalize what designs are bad, and test for them

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Outline of the ER Model

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ER model -- Database Modeling

- The ER data model was developed to facilitate database design by allowing specification of an **enterprise schema** that represents the overall logical structure of a database.
- The ER data model employs three basic concepts:
 - entity sets,
 - relationship sets,
 - attributes.
- The ER model also has an associated diagrammatic representation, the **ER diagram**, which can express the overall logical structure of a database graphically.

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Entity Sets

- An **entity** is an object that exists and is distinguishable from other objects.
 - Example: specific person, company, event, plant
- An **entity set** is a set of entities of the same type that share the same properties.
 - Example: set of all persons, companies, trees, holidays
- An entity is represented by a set of attributes; i.e., descriptive properties possessed by all members of an entity set.
 - Example:


```
instructor = (ID, name, salary)
course = (course_id, title, credits)
```
- A subset of the attributes form a **primary key** of the entity set; i.e., uniquely identifying each member of the set.

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Entity Sets -- *instructor* and *student*

76766	Crick
45565	Katz
10101	Srinivasan
98345	Kim
76543	Singh
22222	Einstein

instructor

98988	Tanaka
12345	Shankar
00128	Zhang
76543	Brown
76653	Aoi
23121	Chavez
44553	Peltier

student

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Representing Entity sets in ER Diagram

- Entity sets can be represented graphically as follows:
 - Rectangles represent entity sets.
 - Attributes listed inside entity rectangle
 - Underline indicates primary key attributes

<i>instructor</i>
<u>ID</u>
name
salary

<i>student</i>
<u>ID</u>
name
tot_cred

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Relationship Sets

- A **relationship** is an association among several entities
 - Example:

44553 (Peltier)	<u>advisor</u>	22222 (Einstein)
<i>student</i> entity	relationship set	<i>instructor</i> entity
- A **relationship set** is a mathematical relation among $n \geq 2$ entities, each taken from entity sets

$$\{(e_1, e_2, \dots, e_n) \mid e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\}$$

where $(e_1, e_2, \dots, e_n)$ is a relationship

- Example:

$$(44553, 22222) \in \text{advisor}$$

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Relationship Sets (Cont.)

- Example: we define the relationship set *advisor* to denote the associations between students and the instructors who act as their advisors.
- Pictorially, we draw a line between related entities.

instructor

student

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Representing Relationship Sets via ER Diagrams

- Diamonds represent relationship sets.

instructor

student

advisor

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Relationship Sets (Cont.)

- An attribute can also be associated with a relationship set.
- For instance, the *advisor* relationship set between entity sets *instructor* and *student* may have the attribute *date* which tracks when the student started being associated with the advisor.

instructor

student

date

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Relationship Sets with Attributes

instructor

student

advisor

date

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Roles

- Entity sets of a relationship need not be distinct
 - Each occurrence of an entity set plays a "role" in the relationship
- The labels "*course_id*" and "*prereq_id*" are called **roles**.

course

prereq

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Degree of a Relationship Set

- Binary relationship
 - involve two entity sets (or degree two).
 - most relationship sets in a database system are binary.
- Relationships between more than two entity sets are rare. Most relationships are binary. (More on this later.)
 - Example: *students* work on research *projects* under the guidance of an *instructor*.
 - relationship *proj_guide* is a ternary relationship between *instructor*, *student*, and *project*

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Non-binary Relationship Sets

- Most relationship sets are binary
- There are occasions when it is more convenient to represent relationships as non-binary.
- E-R Diagram with a Ternary Relationship

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Complex Attributes

- Attribute types:
 - Simple and composite attributes.
 - Single-valued and multivalued attributes
 - Example: multivalued attribute: *phone_numbers*
 - Derived attributes
 - Can be computed from other attributes
 - Example: age, given date_of_birth
- Domain – the set of permitted values for each attribute

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Composite Attributes

- Composite attributes allow us to divided attributes into subparts (other attributes).

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Representing Complex Attributes in ER Diagram

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Mapping Cardinality Constraints

- Express the number of entities to which another entity can be associated via a relationship set.
- Most useful in describing binary relationship sets.
- The cardinality of a relationship is the number of related rows for each of the two objects in the relationship.
- For a binary relationship set the mapping cardinality must be one of the following types:
 - One to one
 - One to many
 - Many to one
 - Many to many

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Mapping Cardinalities

One to one One to many

Note: Some elements in A and B may not be mapped to any elements in the other set

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Reduction to Relation Schemas

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Reduction to Relation Schemas

- Entity sets and relationship sets can be expressed uniformly as *relation schemas* that represent the contents of the database.
- A database which conforms to an E-R diagram can be represented by a collection of schemas.
- For each entity set and relationship set there is a unique schema that is assigned the name of the corresponding entity set or relationship set.
- Each schema has a number of columns (generally corresponding to attributes), which have unique names.

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Representing Entity Sets

- A strong entity set reduces to a schema with the same attributes
student(ID, name, tot_cred)
- A weak entity set becomes a table that includes a column for the primary key of the identifying strong entity set
section (*course_id*, *sec_id*, *sem*, *year*)
- Example



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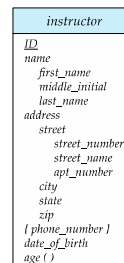
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Representation of Entity Sets with Composite Attributes

- Composite attributes are flattened out by creating a separate attribute for each component attribute
 - Example: given entity set *instructor* with composite attribute *name* with component attributes *first_name* and *last_name* the schema corresponding to the entity set has two attributes *name_first_name* and *name_last_name*
 - Prefix omitted if there is no ambiguity (*name_first_name* could be *first_name*)
- Ignoring multivalued attributes, extended instructor schema is
 - instructor*(ID, first_name, middle_initial, last_name, street_number, street_name, apt_number, city, state, zip, { phone_number }, date_of_birth, age ())



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Representation of Entity Sets with Multivalued Attributes

- A multivalued attribute *M* of an entity *E* is represented by a separate schema *EM*
- Schema *EM* has attributes corresponding to the primary key of *E* and an attribute corresponding to multivalued attribute *M*
- Example: Multivalued attribute *phone_number* of *instructor* is represented by a schema:
inst_phone = (ID, *phone_number*)
- Each value of the multivalued attribute maps to a separate tuple of the relation on schema *EM*
 - For example, an *instructor* entity with primary key 22222 and phone numbers 456-7890 and 123-4567 maps to two tuples: (22222, 456-7890) and (22222, 123-4567)

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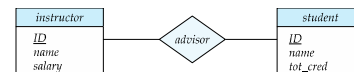
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Representing Relationship Sets

- A many-to-many relationship set is represented as a schema with attributes for the primary keys of the two participating entity sets, and any descriptive attributes of the relationship set.
- Example: schema for relationship set *advisor*

advisor = (s_id, i_id)



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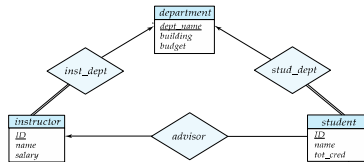
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Redundancy of Schemas

- Many-to-one and one-to-many relationship sets that are total on the many-side can be represented by adding an extra attribute to the "many" side, containing the primary key of the "one" side
- Example: Instead of creating a schema for relationship set *inst_dept*, add an attribute *dept_name* to the schema arising from entity set *instructor*
- Example



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Redundancy of Schemas (Cont.)

- For one-to-one relationship sets, either side can be chosen to act as the "many" side
 - That is, an extra attribute can be added to either of the tables corresponding to the two entity sets
- If participation is *partial* on the "many" side, replacing a schema by an extra attribute in the schema corresponding to the "many" side could result in null values

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Redundancy of Schemas (Cont.)

- The schema corresponding to a relationship set linking a weak entity set to its identifying strong entity set is redundant.
- Example: The *section* schema already contains the attributes that would appear in the *sec_course* schema



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UML

- UML:** Unified Modeling Language
- UML has many components to graphically model different aspects of an entire software system
- UML Class Diagrams correspond to E-R Diagram, but several differences.

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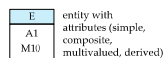
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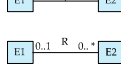
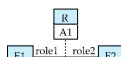
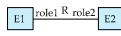
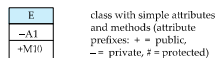


ER vs. UML Class Diagrams

ER Diagram Notation



Equivalent in UML



* Note reversal of position in cardinality constraint depiction

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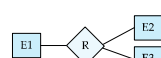
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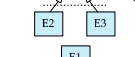
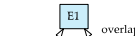


ER vs. UML Class Diagrams

ER Diagram Notation



Equivalent in UML



* Generalization can use merged or separate arrows independent of disjoint/overlapping

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UML Class Diagrams (Cont.)

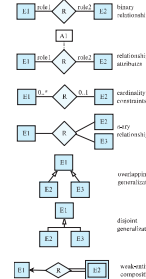
- Binary relationship sets are represented in UML by just drawing a line connecting the entity sets. The relationship set name is written adjacent to the line.
- The role played by an entity set in a relationship set may also be specified by writing the role name on the line, adjacent to the entity set.
- The relationship set name may alternatively be written in a box, along with attributes of the relationship set, and the box is connected, using a dotted line, to the line depicting the relationship set.



ER vs. UML Class Diagrams

ER Diagram Notation

E	entity with attributes (simple, composite, multivalued, derived)
A1	
M10	



Equivalent in UML

E	class with simple attributes and methods (abstract, public, private, protected)
A1	
M10	

